SHAHID MUMTAZ

AI Engineer | Python Developer

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Summary

I'm **AI Engineer** with **3+ years** of specialized experience in **designing**, **developing,** and **deploying** production-ready AI solutions using **Python**. Expertise in **full-stack AI application development**, including **Generative AI**, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), **Graph RAG**, **NLP**, fine-tuning, and agentic systems. Proven track record in building scalable AI pipelines with tools like **LangChain, LangGraph,** Hugging Face, TensorFlow, **FastAPI** and **scikit-learn**, integrated with cloud platforms (**AWS, Azure**) and DevOps practices (**Docker, Kubernetes**). Delivered **end-to-end AI projects** that save time equivalent to three software engineers per year, optimize **resource utilization by 30-50%,** reduce costs through automation (e.g., preventing $50K+ in losses), and support full product lifecycles from **ideation to launch and monitoring**.

Skills

AI & Machine Learning:

Python (Expert), TensorFlow, PyTorch, Hugging Face, Large Language Models (LLMs), LangChain, LangGraph, CrewAI Agentic RAG, Natural Language Processing (NLP), Generative AI, Fine-Tuning, Embeddings, Exploratory Data Analysis (EDA), Data Visualization (Seaborn, Matplotlib), Web Scraping (Crawl4AI), Transformer's, GANN's, CNN, RNN, MCP

Data & Deployment:

SQL (PostgreSQL, MySQL), MongoDB, Vector Databases, Data Pipelines, MLOps, Cloud Platforms (AWS, Azure, GCP), CI/CD (Docker, Kubernetes, Git), Redis, Apache Kafka, Grafana, ETL/Orchestration (n8n), WebRTC (LiveKit), STT/TTS, telephony (Twilio, SIP), CRM Integration

Development & Tools:

FastAPI, Node.js, Next JS, JavaScript/TypeScript, SpringAI (Java)

Experience

META ISOL — Chichawatni, Pakistan

AI Engineer — July 2023 – Present (Freelance)

Led AI/ML development for international clients, **delivering scalable solutions** that enhanced efficiency by 25% on average, saving time equivalent to three software engineers annually through automation and optimized **resource utilization**.

- Designed and deployed **end-to-end AI/ML pipelines** using Python, Hugging Face, LangGraph, and **Agentic RAG**, integrating with **FastAPI** and cloud platforms (AWS, Azure) to handle 20K+ daily queries with 95% accuracy in fraud detection, reducing manual tasks by 50% and operational costs.
- Developed **Generative AI features**, including cloned voice calling systems with NLP and **ElevenLabs integration, automating lead generation** for cold marketing and achieving 35% conversion uplift while supporting **full product launch and monitoring**.
- Built Pakistan's first **AI life insurance premium predictor** using scikit-learn models, analyzing 50K+ data points for 92% prediction accuracy, optimizing resource use and saving client money through predictive insights.
- Created AI-powered **fraud detection and payment platforms** with TensorFlow and data pipelines, preventing potential losses of $50K+ by fine-tuning models for KYC and transaction monitoring.
- Collaborated with business teams to **translate requirements into AI solutions**, including **NLP-based chatbots** and recommendation systems, ensuring scalable deployments with **Docker/Kubernetes** for 99.9% uptime.
- Created Real Estate Properties Applications using modern Technology with **CRM Automation**.
- Designed a **Cloned calling System** for cold marketing to automate leads generation.

N SOFTWARE DEVELOPMENT HOUSE (SMC-PVT.) LTD — Sahiwal, Pakistan

Software Developer — Feb/2021 - June/2023

Led software solutions and IT training for 50+ students, focusing on real-world projects in Python and web development.

- **Instructed 50+ students** in Python, PHP, C++, HTML/CSS, and JavaScript; mentored on e-commerce and recruitment projects, resulting in 80% completion rate and 15 live deployments.
- Created National Level CMS platforms using PHP and MYSQL, supporting 100+ users with **secure authentication** via OAuth2 and JWT, reducing login errors by 70%.
- Implemented **API integrations** for web applications, handling 5K+ daily requests with 98% reliability and enhancing data flow efficiency.

Life Line Family Clinic — Chichawatni, Pakistan

Facebook Manager (Clerk) — Jul/2019 - Sep/2020

Manage Facebook Marketing Compiegne's, working as Chat Agent, dealing with Customers, and Handling Documentation about Accounts.

- **Experience with Taxation:** along Facebook management also Manage Billing and Annually Tax Return and account management.

Projects

LeadFlow Central (SAAS PLATFORM) — Founder | Ongoing

AI-driven marketing SaaS for Local service providers, built with Python, **FastAPI, LangChain**, GraphRAG, Next.js, and Node.js.

- Implemented **Generative AI** for lead extraction, **voice-cloned calling (NLP)**, and **scam detection analytics**, automating 70% of marketing tasks, **saving time equivalent to three engineers annually**, and reducing costs via scalable **Docker/Kubernetes** architecture (99% deployment reliability).
- Deployed multilingual **LLM-based systems for conversation analysis**, **order tracking**, **vendor management**, and **notifications, processing real-time data** with 99% reliability from ideation to production launch.
- Optimized resources by 30-50% **through AI-recommended leads** and **automated workflows**, raising funding potential with modular, scalable design.
- MVP

TTS SIRBULAND GPT (E-Commerce Market Analyzer) — Developer | 2025

AI-powered platform for e-commerce intelligence, using LangChain, LangGraph, Tavily (real-time web search) and FastAPI backend.

- Developed **RAG-integrated** features for **market gap analysis**, **trend detection**, and **competitor intelligence**, processing 10K+ datasets with 88% accuracy improvement over baseline models, saving analysis time and resources.
- Supported full product lifecycle from **EDA to production launch**, optimizing resource utilization by **automating data workflows** and ensuring scalability.
- Delivered end-to-end insights, **enhancing decision-making** for clients in USA/UK/UAE.
- Link | GitHub

PayVibe (AI-Powered Fraud Detection) — Developer | 2025

Secure payment system with **ML-driven fraud prevention**, built with Pandas, scikit-learn, and **QuickBooks/Stripe** integrations.

- Developed AI models for **fraud detection**, KYC, and cash flow optimization, **reducing false positives** by 40% across 5K transactions, preventing $50K+ in losses, and **saving time equivalent** to multiple engineers through automated pipelines.
- Designed real-time monitoring dashboard and automated accounts payable with **vendor scoring**, enhancing security for 50+ vendors and optimizing resources by 30%.
- Managed **end-to-end deployment** from development to production, ensuring robust scalability and cost efficiency.
- Link | GitHub

Investor Properties NY (Real Estate Web App) — Developer | 2025

Automated platform for **property sales/purchases** with AI support.

- **Integrated CRM** and analytics for business decisions; added **customer support chatbot** linked to knowledge base via Chabot.
- Processed 1K+ listings with 90% automation rate using Node.js and real time database Supabase. End-to-end deployment.
- Link | GitHub

Interectors (AI-Powered Chrome Extension) — Developer | 2025

Page Summarizer and Q/A on current page. Using RAG method with LangChain.

- Get concise summaries of any web page.
- Ask questions about the current page content, increasing understanding by 50%. Managed **end-to-end deployment** from development to production.
- Link

Travel Voice Assistant Agent (WebRTC) — Developer | 2025

I developed a full-stack Travel Assistant Agent using LiveKit, Next.js, FastAPI, LangChain, LangGraph, Gemini, and SerpAPI—a complete voice-enabled travel assistant application

- Its provides a sophisticated travel planning experience through natural voice conversations, combining real-time communication with AI-powered travel recommendations, including barge-in handling, dialogue-state recovery, and API connections for bookings and payments.
- Link

Education

Virtual University of Pakistan — Chichawatni, Pakistan
BS Software Engineering — 04/2020 - 08/2025

Certificates

1. **Certificate in Machine Learning Course – INNOVEST RAWAL**
2. **Diploma in Computer Application – Chichawatni College Of Technology (Pakistan)**
3. **Data Science & Machine Learning (Internship Demo) – Edureka**
4. **Introduction to Cyber Security – CISCO**
5. **Soft Skills Communication Guide Training –OEC Pakistan**

Key Achievements

- Founded LeadFlow Central SaaS, delivering AI marketing solutions with pilot partnerships for 20+ providers, automating tasks to save time equivalent to three engineers/year and optimizing resources/costs